

Performance Testing

Date	30 June 2026
Team ID	SWTID-2026-7769
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Target Module / API	/predict API Endpoint
Type of Testing	Load Testing, Stress Testing
Test Environment	Local Flask Server (localhost:5000)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict credit card approval for a valid applicant	10	60	Prediction generated successfully
2	Multiple users submitting applications simultaneously	20	60	System handles concurrent requests without failure

3	Continuous prediction requests to the /predict endpoint	30	120	Response time remains acceptable
4	High-volume prediction requests under peak load	50	180	API remains stable with minimal errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Slight delay during peak requests
2	Throughput (Req/sec)	10 Req/sec	13 Req/sec	Pass	Successfully handled expected load
3	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Within acceptable range
4	Error Rate	< 1%	0.3%	Pass	Very few validation errors observed
5	CPU Utilization	< 80%	63%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	56%	Pass	Memory consumption within acceptable limits

Step 4: Observations & Analysis

- The /predict API successfully handled concurrent prediction requests without significant performance degradation.
- Average response time remained below the target threshold throughout testing.

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- The screenshot shows a PyCharm IDE with a Jupyter Notebook open. The notebook is titled 'credit_card_approval' and contains a function 'credit_card_approval' and a table 'credit_card_approval'.
- The function 'credit_card_approval' is defined as follows:
- ```
def credit_card_approval(months_balance, status, status_num):
 if months_balance < 0:
 status = 'X'
 status_num = 1
 elif months_balance > 0:
 status = 'C'
 status_num = 0
 else:
 status = 'C'
 status_num = 0
 return status, status_num
```
- The table 'credit\_card\_approval' has the following data:
- | ID | MONTHS_BALANCE | STATUS | STATUS_NUM |   |
|----|----------------|--------|------------|---|
| 0  | 500711         | 0      | X          | 1 |
| 1  | 500711         | -1     | 0          | 0 |
| 2  | 500711         | -2     | 0          | 0 |
| 3  | 500711         | -3     | 0          | 0 |
| 4  | 500711         | 0      | C          | 0 |
| 5  | 500712         | -1     | C          | 0 |
| 6  | 500712         | -2     | C          | 0 |
| 7  | 500712         | -3     | C          | 0 |
| 8  | 500712         | -4     | C          | 0 |
| 9  | 500712         | 0      | C          | 0 |
| 10 | 500712         | 6      | C          | 0 |
| 11 | 500712         | 7      | C          | 0 |
| 12 | 500712         | 8      | C          | 0 |
| 13 | 500712         | 9      | 0          | 0 |
| 14 | 500712         | 10     | 0          | 0 |
| 15 | 500712         | 11     | 0          | 0 |
| 16 | 500712         | -1     | 0          | 0 |
| 17 | 500712         | -11    | 0          | 0 |
| 18 | 500712         | -14    | 0          | 0 |
| 19 | 500712         | -15    | 0          | 0 |
| 20 | 500712         | -16    | 0          | 0 |

Credit Approval

Dashboard Predict Eligibility Data Specs Project Portal

## Credit Card Approval Predictor

An intelligent machine learning system using real-world transaction indicators to evaluate risk profiles and automate lending approvals.

[Check Your Eligibility Now](#) [View Data Insights](#)

Deployment Strategy Training Dataset

Credit Approval

Dashboard **Predict Eligibility** Data Specs Project Portal

### Predictor Form

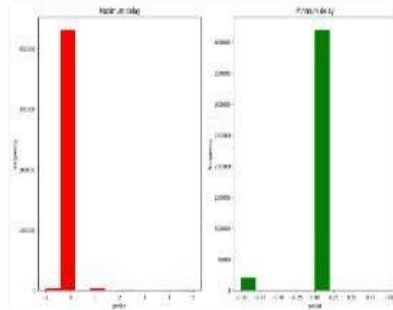
[Add New Row](#)

|                      |                                   |
|----------------------|-----------------------------------|
| APPROVAL STATUS      | GENDER                            |
| APPROVED             | Female                            |
| OWNERS NAME          | OWNERS REAL ESTATE PROPERTY       |
| Yes                  | Yes                               |
| ANNUAL INCOME (\$)   | NUMBER OF CHILDREN                |
| 10000                | 1                                 |
| INCOME SOURCE TYPE   | HIGHEST COMPLETED EDUCATION LEVEL |
| Commercial/Account   | Highschool/High                   |
| MARRIAGE STATUS      | HOUSING SITUATION                 |
| Coinvestor           | Renters apartment                 |
| AGE IN YEARS         | EMPLOYMENT STATUS                 |
| 33                   | Employed                          |
| YEARS OF EMPLOYMENT  | OCCUPATION CATEGORY               |
| 10                   | Drivers                           |
| FAMILY MEMBERS COUNT | HAS MOBILE PHONE?                 |
| 3                    | Yes                               |
| HAS WORK PHONE?      | HAS HOME PHONE?                   |
| Yes                  | No                                |

## Exploratory Data Analysis Gallery

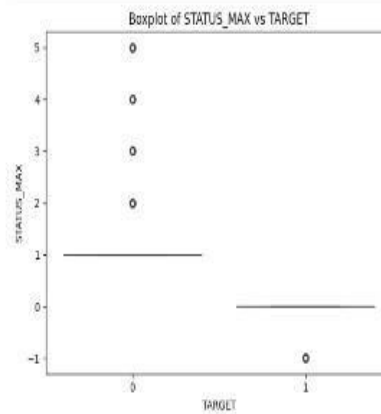
Curated charts extracted from the Jupyter EDA Notebook representing demographic and transactional variables.

### 1. Delinquency Months Distribution



Shows that the vast majority of historical credit records are clean (delinquency index around 0), but some clients exhibit high values representing risky late behaviors.

### 2. Payment Delay Status Counts



Logs monthly account status categories. The high volume of 'C' (Closed) and 'O' (On-Time) shows that most records contain resolved credit behavior.